EVERY POWER TORSION MODULE ADMITS A RESOLUTION BY DIRECT SUMS OF QUOTIENTS OF THE RING BY POWERS OF THE IDEAL

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Editorial note. The proof of the preceding discussion at source lines 23997-23999, characterising power torsion modules as those equal to the union of their submodules annihilated by powers of I, lies outside the counted spans and is not part of this card.

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Allowed standard prerequisites (editorial).

- Every element of an I-power torsion module is annihilated by some power of I, so the smallest such exponent exists. used at proof line 24018, where n_m is defined as the smallest positive integer with I^{n_m} m = 0. Allows the individual summands of the surjection to be chosen with the exponent adapted to each element.
- A direct sum of modules each annihilated by a power of I is again I-power torsion, and a submodule and a quotient of an I-power torsion module are I-power torsion. used at proof lines 24014-24019, where the canonical surjection has power torsion source and its kernel is asserted to be power torsion. Makes the induction run: the kernel of the displayed surjection satisfies the hypothesis of the lemma again.
- A surjection onto M with power torsion source whose kernel is power torsion produces the first two terms of the resolution, and the construction can be iterated. used at proof lines 24019-24020. Turns the single surjection into the infinite resolution of the statement by repeating the construction on the kernel.

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Lemma 0.2. *Let R be a ring. Let I be an ideal of R. Let M be an I-power torsion module. Then M admits a resolution*

$$\ldots \to K_2 \to K_1 \to K_0 \to M \to 0$$

with each K_i a direct sum of copies of R/I^n for n variable.

Proof. There is a canonical surjection

$$\oplus_{m \in M} R/I^{n_m} \to M \to 0$$

where n_m is the smallest positive integer such that $I^{n_m} \cdot m = 0$. The kernel of the preceding surjection is also an I-power torsion module. Proceeding inductively, we construct the desired resolution of M. $\square$